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ELECTRONIC APPARATUS AND SLIDE MECHANISM

Abstract

An electronic apparatus includes: a slider that is movable between a first position and a second position; a shield cover on the slider, the shield cover being located to cover the front of the camera when the slider is in the first position and located to be away from the front of the camera when the slider is in the second position; a movable magnet at the slider; and a fixed magnet at the chassis. The fixed magnet has four poles of magnets. The fixed magnet is configured so that the width of the magnets at both ends is narrower than the width of the center magnets, and the center magnets have the same width and are wider than the width of the magnets at both ends. The width of the movable magnet is equal to or less than half of the width of the fixed magnet.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an electronic apparatus and a slide mechanism.

Description of the Related Art

[0002] Many electronic apparatuses, such as a laptop personal computer (laptop PC), a tablet computer (tablet PC), and a multifunctional mobile phone (smartphone), have optical devices such as a camera around the display.

[0003] Such a camera on the display chassis is typically directed to a user, and the user may feel uneasy because the camera may shoot them contrary to their intention even when the camera does not operate. When the camera is exposed, this may degrade the integrity with the frame in terms of the design. As one means of covering the camera, a manually slidable mechanical shutter including a shield cover can be used as in Japanese Unexamined Patent Application Publication No. 2023-006941.

[0004] The electronic apparatus described in Japanese Unexamined Patent Application Publication No. 2023-006941 includes a frame surrounding a display, and a camera disposed at the upper frame. The shield cover is slidable along the upper frame between a first position and a second position. The shield cover in the first position covers the camera, and the shield cover in the second position exposes the camera.

[0005] This electronic apparatus includes the movable magnet and the fixed magnet that attract to each other when the slider is in the first and second positions. This apparatus does not have a mechanical elastic material, meaning that it is free from influences from component variations or mechanical fatigue. Thus it provides a stable clicking sensation and achieves a long life.

SUMMARY OF THE INVENTION

[0006] Some of the cameras mounted on recent electronic apparatuses have a wide viewing angle, and a shutter that covers such a camera needs to slide a greater distance, but there is a limit to the lateral that width can be accommodated in the shutter mechanism.

[0007] In view of the above, the present invention aims to provide an electronic apparatus and a slide mechanism that are capable of achieving a larger sliding amount while reducing the width of the entire mechanism.

[0008] To solve the above problems, an electronic apparatus according to the first aspect of the present invention includes: a chassis having a display; an optical device disposed in the chassis to direct forward; a slider that is movable between a first position and a second position relative to the chassis; a shield cover in the slider, the shield cover being located to cover the front of the optical device when the slider is in the first position and located to be away from the front of the optical device when the slider is in the second position; a movable magnet at the slider; and a fixed magnet at the chassis, the fixed magnet having four poles of N poles and S poles in total alternately disposed in a moving direction of the slider on a face opposing the movable magnet, the movable magnet having one first magnetic pole and one second magnetic pole disposed in the moving direction on a face opposing the fixed magnet, the first magnetic pole and the second magnetic pole being disposed in order so as to attract to the fixed magnet when the slider is in the first position and the second position, two poles of the fixed magnet at both ends are narrower than central two poles in the moving direction, the N pole and the S pole of the movable magnet having a same width in the moving direction and being wider than the two poles at both ends of the fixed magnet, the total width of the movable magnet being equal to or less than half of an overall width of the

fixed magnet.

[0009] A slide mechanism according to the second aspect of the present invention includes: a slider that is movable between a first position and a second position relative to a chassis; a movable magnet at the slider; and a fixed magnet at the chassis, the fixed magnet having four poles of N poles and S poles in total alternately disposed in a moving direction of the slider on a face opposing the movable magnet, the movable magnet having one first magnetic pole and one second magnetic pole disposed in the moving direction on a face opposing the fixed magnet, the first magnetic pole and the second magnetic pole being disposed in order so as to attract to the fixed magnet when the slider is in the first position and the second position, two poles of the fixed magnet at both ends are narrower than central two poles in the moving direction, the N pole and the S pole of the movable magnet having a same width in the moving direction and being wider than the two poles at both ends of the fixed magnet, the total width of the movable magnet being equal to or less than half of an overall width of the fixed magnet.

[0010] The above-described aspects of the present invention achieve a larger slide amount while suppressing the width of the entire mechanism.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a perspective view of an electronic apparatus according to one embodiment of the present invention.

[0012] FIG. 2 is a cross-sectional perspective view of an upper portion of the display chassis including a groove.

[0013] FIG. 3 is an enlarged front view of the upper frame in the display chassis when the slider is in the first position.

[0014] FIG. 4 is an exploded front view of the upper frame and the slider.

[0015] FIG. 5 is a cross-sectional perspective view of an upper portion of the display chassis including the second guide protrusion piece.

[0016] FIGS. 6A-6B illustrate the movable magnet, the fixed magnet and their peripheral, where FIG. 6A shows them when the slider is in the first position, and FIG. 6B shows them when the slider is in the second position.

[0017] FIGS. 7A-7B schematically illustrate the movable magnet and the fixed magnet that face each other; where FIG. 7A shows them when the slider is in the first position, and FIG. 7B shows them when the slider is in the second position.

[0018] FIG. 8 is a cross-sectional perspective view of an upper portion of the display chassis including the fixed magnet and the movable magnet.

DETAILED DESCRIPTION OF THE INVENTION

[0019] The following describes an electronic apparatus and a slide mechanism according to one embodiment of the present invention in details, with reference to the drawings. The present invention is not limited to the following embodiment.

[0020] FIG. 1 is a perspective view of an electronic apparatus **10** according to one embodiment of the present invention. The electronic apparatus **10** includes a slide mechanism **12** according to one embodiment of the present invention. The electronic apparatus **10** is a laptop PC. The electronic apparatus according to the present invention is not limited to a laptop PC, which may be a desktop PC, a mobile tablet terminal, or a multifunctional mobile phone, for example.

[0021] The electronic apparatus **10** includes a body chassis **14** and a display chassis (chassis) **16** that is openable/closable relative to the body chassis **14** via hinges **18**. The laptop PC with the display chassis **16** closed is compact and is suitable for mobile use. The upper face of the body chassis **14** is provided with a keyboard **20**.

[0022] The following may refer to the left-right direction of the display chassis **16** as X direction, the up-down direction as Y direction, and the thickness direction of a display **22** as Z direction. For the display chassis **16**, lower refers to the side connecting to the body chassis **14** via the hinges **18**, and upper refers to the opposite side. X direction is the moving direction of a slider **36** described later. Y direction is the direction orthogonal to the moving direction of the slider **36** described later and the front-rear direction. For Z direction, the front refers to the direction where the front face of the display **22** faces, and the rear refers to the opposite. That is, Z direction is the front-rear direction.

[0023] The display chassis **16** includes the display **22** on the front face, i.e., on the display face, a frame (chassis) **24** surrounding the display **22**, and a rear cover (chassis) **26**. The frame **24** and the rear cover **26** make up a part of the chassis. The rear cover **26** and the frame **24** are made of resin materials. The rear cover **26** has a slightly larger area than the display **22** and covers the rear face of the display **22**. The display **22** is a rectangular, thin color liquid crystal display, and occupies a major part of the front face of the display chassis **16**. The frame **24** is narrow, which may be called a narrow bezel.

[0024] FIG. **2** is a cross-sectional perspective view of an upper portion of the display chassis **16** including a groove **31a**. As illustrated in FIGS. **1** and **2**, a camera (optical device) **28** and a camera lamp (optical device) **29** are placed at the rear cover **26** in a position behind an upper frame **24a** of the frame **24** so that the camera **28** and the camera lamp **29** face forward.

[0025] The display chassis **16** has a glass **34** on the front face. The glass **34** covers the front portion of the display **22** and of the upper frame **24a**. The portion of the glass **34** that covers the upper frame **24a** is colored in black so that the structure of the upper frame **24a** is invisible. The portion of the glass **34** that is located in front of the camera **28** and the camera lamp **29** has no color and serves a transparent window to expose these devices. The camera **28** and the camera lamp **29** may be shielded by a first shield cover **46a** and a second shield cover **46b** described later. The camera **28** is placed at a substantially center position of the upper frame **24a** in the left-right direction. A microphone and an infrared port may be disposed at the upper frame **24a**.

[0026] A narrow edge wall **30** surrounds the circumference of the rear cover **26**. A narrow circumferential wall **31** surrounds almost the entire circumference of the frame **24**. The circumferential wall **31** is substantially in contact with the inner circumference of the edge wall **30**. The circumferential wall **31** protrudes slightly from the front edge of the edge wall **30**.

[0027] The circumferential wall **31** has a groove **31a** at a substantially center of the upper frame **24a**. The groove **31a** is provided with an operation knob **32**. The operation knob **32** is narrower in X direction than the groove **31a**. The operation knob **32** and the circumferential wall **31** have the same width in Y direction. The operation knob **32** protrudes forward from between the edge wall **30** and the glass **34** to the same extent as the circumferential wall **31** protrudes. The operation knob **32** is small enough to fit within the width of the circumferential wall **31** in Y direction and Z direction and be less conspicuous, and is favorable in appearance.

[0028] The operation knob **32** is a part of the slider **36** (see FIG. **3**) described later, and slides in X direction together with the slider **36**. In other words, the slider **36** is movable in X direction relative to the display chassis **16** between a first position and a second position. The operation knob **32** is slidable between the first position (see FIG. **6A**) and the second position (see FIG. **6B**) within the width of the groove **31a**. The operation knob **32** has a cutout **32a**, which allows the user to operate the operation knob **32** easily with a finger or a nail. The cutout **32a** is located just above the camera **28** when the slider **36** is in the first position, which enables the user to operate the operation knob **32** intuitively to cover the camera **28**.

[0029] FIG. **3** is an enlarged front view of the upper frame **24a** in the display chassis **16** when the slider **36** is in the first position. FIG. **3** omits the glass **34** so that the structure of the upper frame **24a** is visible. FIG. **4** is an exploded front view of the upper frame **24a** and the slider **36**.

[0030] As illustrated in FIG. **3** and FIG. **4**, the upper frame **24a** is provided with the slider **36**. The

slider **36** is movable in the direction along the edge wall **30** (i.e., in X direction). When the slider **36** moves to the leftmost in the front view, the operation knob **32** abuts a first contact portion **31aa** at the left end of the groove **31a** to be restricted in movement to the left (see FIG. 3). This position of the slider **36** is referred to as the first position. When the slider **36** is displaced to the rightmost in the front view, the operation knob **32** abuts a second contact portion **31ab** at the right end of the groove **31a** to be restricted in movement to the right. This position of the slider **36** is referred to as the second position. In this embodiment, the first position is on the left in X direction (one side along the moving direction), and the second position is on the right (the other side along the moving direction).

[0031] The slider **36** includes a metal plate **38** that is the base. The metal plate **38** is made of a ferrite material. The metal plate **38** has a substantially rectangular main portion **38a** on the left and an arm **38b** protruding from the lower part of the main portion **38a** to the right. The upper left portion of the main portion **38a** is integral with the operation knob **32**. A first guide protrusion piece **39** (see FIG. 2) and a first slide piece **40** (see FIG. 2) are placed behind the operation knob **32**. The upper right portion of the main portion **38a** is integral with a second slide piece **41** (see FIG. 5) protruding rearward and a second guide protrusion piece **42** protruding upward from the rear end of the second slide piece **41** (see FIG. 5).

[0032] The main portion **38a** has a horizontally long exposure hole **44** in a substantially center. A portion of the main portion **38a** on the left of the exposure hole **44** is a first shield cover **46a**, and a portion on the right is a second shield cover **46b**. When the slider **36** is in the first position (see FIG. 3), the first shield cover **46a** is positioned to cover the front of the camera **28** and the second shield cover **46b** is positioned to cover the front of the camera lamp **29**. When the slider **36** is in the second position, the first shield cover **46a** is positioned away from the front of the camera **28** and the second shield cover **46b** is positioned away from the front of the camera lamp **29**. When the slider **36** is in the second position, the camera lamp **29** is exposed through the exposure hole **44**.

[0033] Referring back to FIG. 2, the first guide protrusion piece **39** protrudes slightly rearward while being continuous with the operation knob **32**. The rear cover **26** has a guide pedestal **47** at the corner of the edge wall **30** and the bottom face **43**. The guide pedestal **47** has a protrusion piece **47a** that slightly protrudes forward from the lower portion. The protrusion piece **47a** and the edge wall **30** define a first guide groove **49** therebetween.

[0034] The first slide piece **40** is located slightly below the first guide protrusion piece **39** and protrudes rearward in parallel with the first guide protrusion piece **39**. The first slide piece **40** abuts and slides on the bottom face **43**. The first slide piece **40** has an appropriate width in Y direction, and thus allows the slider **36** to be stably supported and be slidable relative to the rear cover **26**. The first guide protrusion piece **39** and the first slide piece **40** define a second guide groove **50** therebetween.

[0035] The first guide protrusion piece **39** fits into the first guide groove **49**, and the protrusion piece **47a** fits into the second guide groove **50**. This configuration restricts the displacement of the slider **36** in Y direction and properly guides the slider **36** in X direction during sliding. In this way, the protrusion piece **47a** serves as a rail guide that guides the slider **36** in X direction.

[0036] Note here that a narrow gap G1 in Y direction is formed between the first guide protrusion piece **39** and the protrusion piece **47a**, and a narrow gap G2 in Y direction is formed between the protrusion piece **47a** and the first slide piece **40**. In this way, the slider **36** is allowed to tilt somewhat. When the slider **36** is in the first position or the second position, the slider **36** is biased upward by the attraction action between a movable magnet **54** and a fixed magnet **56**, which will be described later, so that no gap is formed between the first guide protrusion piece **39** and the edge wall **30**. This gives favorable appearance.

[0037] FIG. 5 is a cross-sectional perspective view of an upper portion of the display chassis **16** including the second guide protrusion piece **42**. The rear face of the second slide piece **41** and the rear face of the second guide protrusion piece **42** define a continuous surface, and abut and slide on

the bottom face **43**. These rear faces of the second slide piece **41** and the second guide protrusion piece **42** have an appropriate area and thus allow the slider **36** to be stably supported and slide relative to the rear cover **26**.

[0038] A part of the upper frame **24a** forms a protrusion piece **24aa** protruding rearward. The protrusion piece **24aa** and the bottom face **43** define a guide gap **52** therebetween. The second guide protrusion piece **42** fits into the guide groove **52**. This configuration restricts the displacement of the slider **36** in Z direction and properly guides the slider **36** in X direction during sliding. In this way, the protrusion piece **24aa** serves as a rail guide that guides the slider **36** in X direction. When the slider **36** is in the first position or the second position, the slider **36** is biased upward by the attraction action between the fixed magnet **56** and the movable magnet **54**, which will be described later, so that the second guide protrusion piece **42** comes in contact with the edge wall **30** and is restricted in displacement upward so as to be stable. Note that the second guide protrusion piece **42** is not restricted in downward displacement.

[0039] Referring back to FIG. 3 and FIG. 4, the arm **38b** is provided with the movable magnet **54** at the right end. A part of the arm **38b** is bent to wrap the movable magnet **54** for holding (see also FIG. 8). The movable magnet **54** is of a square upper and lower double-sided multi-pole magnetized type, and is magnetized so that the N pole and the S pole are disposed from left to right on the upper side to have a two-pole configuration. The movable magnet **54** is magnetized so that the S pole and the N pole are disposed from left to right on the lower side. For convenience, these magnetized parts are referred to as magnets **54a** and **54b** from left to right, but they may be referred to as other names such as magnetic pole parts. In each drawing of the present application, the N poles are illustrated with dots so that the polarity can be easily identified.

[0040] The upper frame **24a** is provided with the fixed magnet **56** at the portion above the range where the movable magnet **54** moves in X direction. That is, the fixed magnet **56** and the movable magnet **54** are placed side by side in Y direction, which suppresses the thickness of the display chassis **16** in Z direction.

[0041] The fixed magnet **56** is held by a part of the upper frame **24a** (see also FIG. 8). The fixed magnet **56** is of a square upper and lower double-sided multi-pole magnetized type, and is magnetized so that the S pole, the N pole, the S pole, and the N pole are disposed from left to right on the lower side to have a four-pole configuration. The fixed magnet **56** is magnetized so that the N pole, the S pole, the N pole, and the S pole are disposed from left to right on the upper side. For convenience, these magnetized parts are referred to as magnets **56a**, **56b**, **56c** and **56d** from left to right, but they may be referred to as other names such as magnetic pole parts.

[0042] That is, the fixed magnet **56** may have four poles of N poles and S poles in total alternately disposed in X direction of on a face opposing the movable magnet **54**. The fixed magnet **54** may have two poles of N pole and S pole in total disposed in X direction of on a face opposing the fixed magnet **56**, that is, these poles are placed in order so that the movable magnet **54** attract to the fixed magnet **56** when the slider **36** is in the first and second positions (see FIG. 6). The movable magnet **54** has two fewer poles than the fixed magnet **56**.

[0043] FIG. 6 schematically illustrates the movable magnet **54**, the fixed magnet **56** and their peripheral; where FIG. 6A shows them when the slider **36** is in the first position, and FIG. 6B shows them when the slider **36** is in the second position.

[0044] The dimensions of the movable magnet **54** and fixed magnet **56** are described below, all of which are just examples. The width L1 of the movable magnet **54** in X direction is 4 mm. The width L2 of the fixed magnet **56** in X direction is 9 mm. The width L10 of the magnets **54a** and **54b** in X direction is 2 mm, which is half of L1. The widths of the magnets **56a** and **56d** at both ends of the fixed magnet **56** are equal and indicated as L21; L21 is 1.4 mm. The widths of the central magnets **56b** and **56c** are equal and indicated as L22; L22 is 3.1 mm. The fixed magnet **56** is formed by magnetizing a single magnetic material, meaning that there is no gap between the magnet **56b** and magnet **56c**, and a strong magnetic force is generated by an appropriately large

width L22.

[0045] In other words, the fixed magnet **56** is configured in X direction so that the two-pole magnets **56a** and **56d** at both ends are narrower than the central two-pole magnets **56b** and **56c**. The movable magnet **54** is configured in X direction so that the magnets **54a** and **54b** have the same width and are wider than the magnets **56a** and **56d**, and the total width L1 is equal to or less than half of the overall width L2 of the fixed magnet **56**. Such a length L1 of the movable magnet **54**, which is set sufficiently shorter than the width L2 of the fixed magnet **56**, keeps its long operating stroke. This also reduces the weight of the slider **36** that is a moving body, which improves the dynamic characteristics of the moving body.

[0046] When the slider **36** is located in the first position (FIG. 6A) or in the second position (FIG. 6B), the fixed magnet **56** and the movable magnet **54** face each other with partially the same poles overlapping. Specifically, in the first position, the N-pole magnet **54a** and magnet **56b** partially overlap, and in the second position, the S-pole magnet **54b** and magnet **56c** partially overlap. The width L3 of these overlapping is narrower than the width L21 of the magnets **56a** and **56d** at both ends of the fixed magnet **56**.

[0047] As illustrated in FIG. 6A, which shows the slider **36** in the first position, the movable magnet **54** and the fixed magnet **56** are placed so that different poles face each other and attract each other. Specifically, a part of the N pole of the magnet **54a** and a part of the S pole of the magnet **56a** face each other and a part of the S pole of the magnet **54b** and a part of the N pole of the magnet **56b** face each other to attract each other. This keeps the slider **36** in the first position. When the slider **36** is in the first position, the first contact portion **31aa** (see FIG. 3) restricts the slider **36** from moving leftward. In this state, the left ends of the movable magnet **54** and of the fixed magnet **56** are positioned to almost coincide with each other.

[0048] As illustrated in FIG. 6B, which shows the slider **36** in the second position, the movable magnet **54** and the fixed magnet **56** are placed so that different poles face each other and attract each other. Specifically, a part of the N pole of the magnet **54a** and a part of the S pole of the magnet **56c** face each other and a part of the S pole of the magnet **54b** and a part of the N pole of the magnet **56d** face each other to attract each other. This keeps the slider **36** in the second position. When the slider **36** is in the second position, the second contact portion **31ab** (see FIG. 3) restricts the slider **36** from moving rightward. In this state, the right ends of the movable magnet **54** and of the fixed magnet **56** are positioned to almost coincide with each other.

[0049] In this way, when the slider **36** is in the first position, the left ends of the fixed magnet **56** and the movable magnet **54** are in the same position, and when the slider **36** is in the second position, the right ends of the fixed magnet **56** and the movable magnet **54** are in the same position. In this way, the movable magnet **54** effectively utilizes the entire length of the width L2 of the fixed magnet **56** to keep its long stroke amount. If the left end of the movable magnet **54** in the first position and the right end of the movable magnet **54** in the second position protrude beyond the range of width L2 of the fixed magnet **56**, the space occupied by the slide mechanism **12** will become large. Thus, it is desirable to keep them within the range of width L2. The stroke amount between the first and second positions of the slider **36** in this embodiment achieves 4.9 mm.

[0050] FIG. 7 schematically illustrates the movable magnet **54** and the fixed magnet **56** that face each other; where FIG. 7A shows them when the slider **36** is in the first position, and FIG. 7B shows them when the slider **36** is in the second position. In FIG. 7, the attractive forces between the S pole and N pole of the movable magnet **54** and the fixed magnet **56** are schematically indicated by arrows.

[0051] As illustrated in FIG. 7A, when the slider **36** is in the first position, the center of the magnet **54a** is slightly shifted to the right in X direction from the center of the magnet **56a**, and the center of the magnet **54b** is slightly shifted to the right in X direction from the center of the magnet **56b**. With this arrangement, the movable magnet **54** and the left half of the fixed magnet **56** are attracted to each other in Y direction, and the slider **36** is biased to the left in X direction so that the

operation knob **32** of the slider **36** is pressed against the first contact portion **31aa**.

[0052] The left ends of the movable magnet **54** and the fixed magnet **56** are positioned to almost coincide with each other. Here, the width **L10** of the magnet **54a** is wider than the width **L21** of the magnet **56a**, and thus the direction of the attractive force between them is moderately and significantly inclined, so that the movable magnet **54** is reliably biased to the left, and the slider **36** will not wobble to the right even if a small external force is applied. Furthermore, the width **L22** of the magnet **56b** is set larger than the width **L10** of the magnet **54b**, creating a strong magnetic force and providing a reasonably strong attractive force between these magnet **56b** and magnet **54b**.

[0053] As illustrated in FIG. 7B, when the slider **36** is in the second position, the center of the magnet **54a** is slightly shifted to the left in X direction from the center of the magnet **56c**, and the center of the magnet **54b** is slightly shifted to the left in X direction from the center of the magnet **56d**. With this arrangement, the movable magnet **54** and the right half of the fixed magnet **56** are attracted to each other in Y direction, and the slider **36** is biased to the right in X direction so that the operation knob **32** of the slider **36** is pressed against the second contact portion **31ab**.

[0054] The right ends of the movable magnet **54** and the fixed magnet **56** are positioned to almost coincide with each other. Here, the width **L10** of the magnet **54b** is wider than the width **L21** of the magnet **56d**, and thus the direction of the attractive force between them is moderately and significantly inclined, so that the movable magnet **54** is reliably biased to the right, and the slider **36** will not wobble to the left even if a small external force is applied. Furthermore, the width **L22** of the magnet **56c** is set larger than the width **L10** of the magnet **54a**, creating a strong magnetic force and providing a reasonably strong attractive force between these magnet **56c** and magnet **54a**.

[0055] In other words, conceptually, the narrow magnets **56a** and **56d** bias the movable magnet **54** in the left-right direction, while the attractive force of these magnets tends to be small. Then, the wide magnets **56c** and **56d**, which have a strong attractive force, compensate for the static upward biasing force and the dynamic biasing force during sliding.

[0056] While the width **L21** of the magnets **56a** and **56d** is relatively narrow, they have a minimum, large width **L3** (see FIG. 6) where the same poles overlap each other on their facing portion. If the width **L3** is excessively wide, the repulsion force between the same poles will increase, reducing the attractive force between different poles. In this embodiment, the width **L3** is limited to be narrower than the width **21**, which keeps an optimal attractive force.

[0057] The movable magnet **54** and the fixed magnet **56** are always separated from each other, and a narrow gap **G3** is formed therebetween when the slider **36** is in the first or second position (see also FIG. 8). This gap acts to bias the slider **36** to be attracted upward by the movable magnet **54**, and to press the second guide protrusion piece **42** against the edge wall **30** (see FIG. 5).

Appropriately setting the gap **G3** keeps the movable magnet **54** and the fixed magnet **56** to be separated from each other even if there is some dimensional error or vibrations during operation.

[0058] In this way, when the slider **36** is in the first position or the second position, the operation knob **32** comes in contact with the first contact portion **31aa** or the second contact portion **31ab** without a gap, and the second guide protrusion piece **42** comes in contact with the edge wall **30** without a gap. The operation knob **32** and the second guide protrusion piece **42** are also appropriately pressed against there to be stable without rattling in X and Y directions. This improves the usability of the apparatus.

[0059] FIG. 8 is a cross-sectional perspective view of an upper portion of the display chassis **16** including the fixed magnet **56** and the movable magnet **54**. As described above, the fixed magnet **56** is held by the upper frame **24a** of a resin material, and the movable magnet **54** is held by the arm **38b** of a ferrite material. The display chassis **16** is thin, and the upper frame **24a** is made of a resin material and needs to have an appropriate thickness from the viewpoint of strength and molding conditions. The fixed magnet **56** therefore has width **H1** in Z direction that is slightly narrow due to these restrictions. The arm **38b** is made of a ferrite material that is a metal plate, and can be made thin while maintaining its strength. The movable magnet **54** is therefore allowed to have an

appropriately large width H2 ($H2 > H1$). That is, while width H1 of the fixed magnet 56 is somewhat small, width H2 of the movable magnet 54 is set to be somewhat large. This keeps the magnetic force (attraction and repulsive forces) between these magnets appropriately large. [0060] The movable magnet 54 has ferrite on the non-attracting face opposite to the attracting face. This prevents magnetic leakage to the non-attracting face while guiding the magnetic flux to the attracting face to increase the magnetic flux density on the attracting face. This increases the attracting force of the movable magnet 54 and provides better feeling. The ferrite material is magnetically attracted, and is attracted to the movable magnet 54 during assembly, which facilitates the assembly. The width Y1 of the fixed magnet 56 in Y direction and the width Y2 of the movable magnet 54 in Y direction may be dimensions that allow mounting in the surrounding area of the display 22 of a typical display chassis 16.

[0061] As illustrated in FIG. 3, when the slider 36 is in the first position, the first shield cover 46a covers the camera 28, and the second shield cover 46b covers the camera lamp 29. As described above, the slider 36 is kept in the first position by the attractive action between the movable magnet 54 and the fixed magnet 56, and is biased to the left and upward so as not to rattle.

[0062] When the camera 28 is used, the user moves the operation knob 32 to the right until the slider 36 moves to the second position. For this operation, the gap G3 is kept between the movable magnet 54 and the fixed magnet 56. This prevents the slider 36 from generating unpleasant frictional resistance at least in this portion, so that the operation knob 32 starts to move smoothly. Note here that the movable magnet 54 and the fixed magnet 56 are attracted to each other at the first position, and this generates an appropriate magnetic resistance that the user feels to move the slider 36, and this resistance increases with the amount of movement.

[0063] When the slider 36 is in the second position, the first shield cover 46a is positioned away from the front of the camera 28 and the second shield cover 46b is positioned away from the front of the camera lamp 29. The camera 28 and the camera lamp 29 then are exposed on the front, so that their functions of the optical devices become effective. As described above, the slider 36 is kept in the second position by the attractive action between the movable magnet 54 and the fixed magnet 56, and is biased to the right and upward so as not to rattle. Movement of the slider 36 from the second position to the first position is the reverse operation of the above, so the description thereof will be omitted.

[0064] While the slider 36 is moving between the first position and the second position (not illustrated), the N pole of the magnet 54a and the N pole of the magnet 56b face each other and the S pole of the magnet 54b and the S pole of the magnet 56c face each other, and they repel each other due to the repulsion force between the same poles facing each other. Thus, the user feels a moderate sense of resistance, and beyond the midpoint, feels the reversal of direction of the acting force due to the attractive force. This gives a user a moderate clicking sensation.

[0065] As described above, the gap G1 is formed between the first guide protrusion piece 39 and the protrusion piece 47a, and the gap G2 is formed between the protrusion piece 47a and the first slide piece 40. Further, the second guide protrusion piece 42 is not restricted in downward displacement. In this way, the slider 36 is allowed to tilt somewhat. With this configuration, when a repulsive force is generated between the movable magnet 54 and the fixed magnet 56, the arm 38b on the right of the slider 36 tilts starting from the main portion 38a on the left, so that the movable magnet 54 and the fixed magnet 56 are further separated from the above-mentioned gap G3. This mitigates the repulsive force acting between the movable magnet 54 and the fixed magnet 56 somewhat, and prevents excessive resistance to move the slider 36. This also prevents excessive friction from being generated between the first guide protrusion piece 39 (see FIG. 2) or the first slide piece 40 and the first guide groove 48 and the second guide groove 50. In the case of the operation knob having the mechanism including the blade spring as in Japanese Unexamined Patent Application Publication No. 2023-006941, the operation knob is slightly tilted by the action of the blade spring during the operation. The present embodiment also involves similar tilting, so

that the user is allowed to operate the operation knob without any discomfort compared to the conventional mechanism.

[0066] As described above, the fixed magnet **56** of the electronic apparatus **10** is configured in X direction so that the two-pole magnets **56a** and **56d** at both ends are narrower than the central two-pole magnets **56b** and **56c**. The movable magnet **54** is configured in X direction so that the magnets **54a** and **54b** have the same width and are wider than the magnets **56a** and **56d** of the fixed magnet **56**. This allows the directions of the attractive forces mainly between the magnets **54a** and **56a** and between the magnets **54b** and **56d** to be moderately inclined, and thus generates an appropriate biasing force to the left and right. This therefore keeps a large sliding amount of the slider **36** while restricting the width of the slide mechanism **12** in X direction. Further, the total width **L1** is equal to or less than half the overall width **L2** of the fixed magnet **56**, meaning that sliding over the entire length of the width **L2** allows a stroke amount of at least $L2/2$ to be kept.

[0067] The present invention is not limited to the above-described embodiments, and can be modified freely without deviating from the scope of the present invention.

Claims

1. An electronic apparatus comprising: a chassis having a display; an optical device in the chassis; a slider that is movable between a first position and a second position relative to the chassis; a shield cover in the slider, the shield cover being located to cover a front of the optical device when the slider is in the first position and located to be away from the front of the optical device when the slider is in the second position; a movable magnet at the slider; and a fixed magnet at the chassis, the fixed magnet having a total of four poles of N poles and S poles, the N poles and the S poles being alternately disposed in a moving direction of the slider on a face opposing the movable magnet, the movable magnet having one first magnetic pole and one second magnetic pole disposed in the moving direction on a face opposing the fixed magnet, the first magnetic pole and the second magnetic pole being disposed in order to attract to the fixed magnet when the slider is in the first position and the second position, two poles of the fixed magnet at both ends thereof are narrower than central two poles in the moving direction, the N pole and the S pole of the movable magnet having a same width in the moving direction and being wider than the two poles at both ends of the fixed magnet, a total width of the movable magnet being equal to or less than half of an overall width of the fixed magnet.
2. The electronic apparatus according to claim 1, wherein a first contact portion contacts the slider when located at the first position to restrict a movement of the slider; and a second contact portion that contacts the slider when located at the second position to restrict the movement of the slider, wherein the movable magnet and the fixed magnet are disposed so that the N pole on one of the magnets and the S pole on the other magnet are displaced in the moving direction of the slider to press the slider, when in the first position, against the first contact portion, and are disposed so that the N pole on one of the magnets and the S pole on the other magnet are displaced in the moving direction of the slider to press the slider, when in the second position against the second contact portion.
3. The electronic apparatus according to claim 1, wherein when the slider is located in the first position or in the second position, the fixed magnet and the movable magnet face each other with the same poles overlapping in a width that is narrower than a width of each of the poles at both ends of the fixed magnet.
4. The electronic apparatus according to claim 1, wherein the first position is one side along the moving direction, and the second position is the other side along the moving direction, when the slider is in the first position, one pair of ends of the fixed magnet and movable magnet is in a same position, and when the slider is in the second position, the other pair of ends of the fixed magnet and movable magnet is in a same position.

5. A slider mechanism comprising: a slider that is movable between a first position and a second position relative to a chassis; a movable magnet at the slider; and a fixed magnet at the chassis, the fixed magnet having a total of four poles of N poles and S poles, the N poles and the S poles being alternately disposed in a moving direction of the slider on a face opposing the movable magnet, the movable magnet having one first magnetic pole and one second magnetic pole disposed in the moving direction on a face opposing the fixed magnet, the first magnetic pole and the second magnetic pole being disposed in order to attract to the fixed magnet when the slider is in the first position and the second position, two poles of the fixed magnet at both ends thereof are narrower than central two poles in the moving direction, the N pole and the S pole of the movable magnet having a same width in the moving direction and being wider than the two poles at both ends of the fixed magnet, a total width of the movable magnet being equal to or less than half of an overall width of the fixed magnet.
